

Workshop: Partial Differentiation

UEA LEARNING ENHANCEMENT TEAM MATHEMATICS & STATISTICS

1. For each of the functions below find

$$\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y \partial x}$$

a. $z = 5x + 2y$

b. $z = 5x^3 + 3y^2 + 4$

c. $z = 8xy$

d. $z = 2xy^{0.5}$

e. $z = e^{2x} + 4e^y$

f. $z = x^2 + 2xy + y^2$

g. $z = 10x^2y^3 - 2x - 4y$

h. $z = 5x^2 + 2x + 5y^3 - 2y^2$

i. $z = 10x^4y^3$

j. $z = 2e^{5x} - 4x^2y^2 + \ln 3xy$

k. $z = 2x^2e^3$

l. $z = x^3 \ln y$

m. $z = y(5 - 2x)$

n. $z = y(5 - 2x)^2$

First Order Conditions

2. For each of the functions $f(x, y)$ given find:

i. f_x and f_y

ii. Find the value of x, y , and $f(x, y)$ where both $f_x = 0$ and $f_y = 0$

a. $f(x, y) = x^2 + y^2$

b. $f(x, y) = x^2 - 2x - y^2 - 4y - 3$

c. $f(x, y) = x^2y - 2xy^2 + 3xy + 4$